

IIIT Bangalore

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Question 13

Question

The complete set of only those logic gates designated as universal gates is

- (a) NOT, OR and AND
- (b) XNOR, NOR and NAND
- (c) NOR and NAND
- (d) XOR, NOR and NAND

Analysis

Universal gates are gates that can implement any Boolean function.

For two inputs A and B

NAND gate output

$$NAND = \overline{A \cdot B}$$

NOR gate output

$$NOR = \overline{A + B}$$

Using only NAND gates or only NOR gates we can construct AND, OR and NOT gates. Therefore NAND and NOR are called universal gates.

Correct option: (c) NOR and NAND.

Python Code

```
import numpy as np
import matplotlib.pyplot as plt
```

```
A = np.array([0,0,1,1])
```

```
B = np.array([0,1,0,1])
```

```
NAND = 1 - (A & B)
```

```

NOR = 1 - (A | B)

print("A B NAND NOR")

for i in range(4):
    print(A[i], B[i], NAND[i], NOR[i])

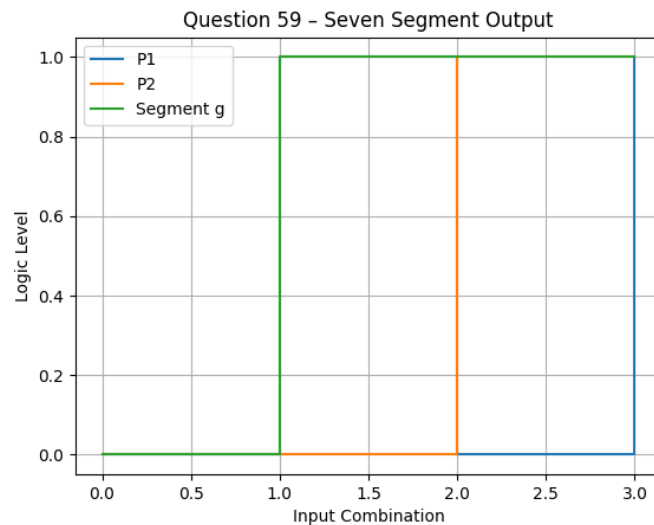
x = np.arange(4)

plt.step(x,A,where='post',label='A')
plt.step(x,B,where='post',label='B')
plt.step(x,NAND,where='post',label='NAND')
plt.step(x,NOR,where='post',label='NOR')

plt.grid()
plt.legend()
plt.savefig("graph.png")
plt.show()

```

Graph



Conclusion

From the truth table and the graph it is clear that NAND and NOR gates can implement all basic logic gates. Hence they are universal gates.